

Show all your work in the space provided for each problem. All problems will be graded for correctness, completeness, conciseness, and clarity. Do your best. Don't cut corners.

1. (8 pts) List all the elements and the binary operation which make each set a group.

a) $\mathbb{Z}_6 =$

operation

b) $U(8) =$

operation

2. (4 pts) Write each of the following statements using the “divides” notation $x|y$

a is a divisor of b

a is a multiple of b

If p is a factor of n and x is any integer then the product nx is divisible by p.

3. (6 pts) Let G be a group with operation *. Use existential and universal quantifiers as needed to define the following properties for G.

a) The associative property

b) The inverse property

c) The commutative property

4. (8 pts) What is the order of each of the following groups?

a) $|Z_n| =$

b) If p is prime $|U(p)| =$

c) $|D_n| =$

d) $|S_n| =$

5. (8 pts) Use the Cayley table given below for the group G with operation $\circ$ to answer the following questions.

a) (2 pt) What is the inverse of A ?

b) (2 pts) List all elements of order 2 in G .

c) (2 pts) List all elements in the center of G .

d) (2 pts) Prove that G is **not** Abelian.

	1	A	B	C	D	E	F	G
1	1	A	B	C	D	E	F	G
A	A	B	C	1	G	F	D	E
B	B	C	1	A	E	D	G	F
C	C	1	A	B	F	G	E	D
D	D	F	E	G	1	B	A	C
E	E	G	D	F	B	1	C	A
F	F	E	G	D	C	A	1	B
G	G	D	F	E	A	C	B	1

6. (12 pts) Let $\alpha = (1\ 3\ 5\ 7\ 9)(2\ 4)(6\ 8)$ and $\beta = (1\ 5\ 2\ 4)(6\ 7\ 8\ 9)$ be permutations in S_9 .

a) Write β in **array form**. $\beta =$

b) Write the product $\alpha\beta$ in **standard cycle form**. $\alpha\beta =$

c) What is the order of each element: $|\alpha| =$ $|\beta| =$

d) Write α^{154} in standard cycle form. $\alpha^{154} =$

7. (6 pts) Shown below is a blank Cayley table for the group D_4 . Write the elements in cells labeled a and b as a product of elements in D_4 . Then draw diagrams to determine exactly which element should be in the cell. You must show your diagrams to receive full credit.

a =

Diagram for a:

b =

Diagram for b:

	R₀	R₉₀	R₁₈₀	R₂₇₀	H	V	D	D'
R₀								
R₉₀								
R₁₈₀								
R₂₇₀								
H								
V		a						
D								b
D'								



H



V



D



D'

8. (8 pts) Let $\sim$ be an equivalence relation on a set S . Prove that any two equivalence classes $[a]$ and $[b]$ are either disjoint or identical (either $[a] \cap [b] = \emptyset$ or $[a] = [b]$)

9. (10 pts)

a) (2 pts) The prime factorization for x is $11^5 * 23^4 * 47^2 * 61$ and the prime factorization for the integer y is $7^3 * 11^2 * 13 * 61^5$. What is the prime factorization for the $\gcd(x,y)$?

$\gcd(x,y) =$

b) (8 pts) Use the Euclidean algorithm to find the $\gcd(589, 323)$ and find integers x and y such that $\gcd(589, 323) = 589*x + 323*y$.

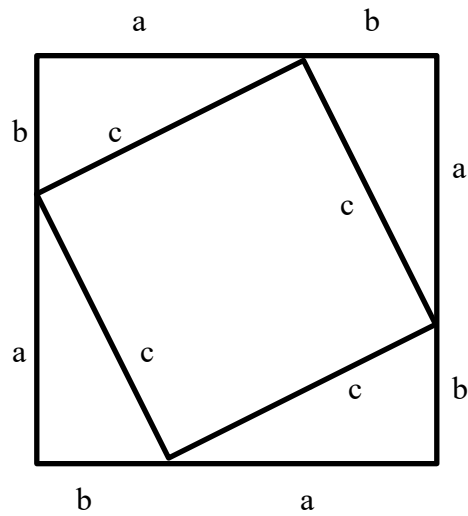
10. (15 pts) Consider $\mathbf{R}^2 = \{ (a,b) : a \text{ and } b \text{ are real numbers} \}$. We can define a relation $\sim$ on $\mathbf{R}^2$ by saying two points (a,b) and (x,y) are related exactly when the difference between the y-coordinates is twice as much as the difference between the x-coordinates. i.e. $(a,b) \sim (x,y)$ if and only if $b-y = 2(a-x)$.

a) (3 pt) List three ordered pairs that are related to $(0,5)$.

b) (9 pts) Prove that $\sim$ is an equivalence relation on the set of all ordered pairs $\mathbf{R}^2$.

c) (3 pts) Give a geometric description of all the points in the equivalence class $[(0,5)]$

11. (6 pts) Explain in complete sentences and equations how this diagram proves the Pythagorean Theorem.



12. (9 pts) Let a and b be any integers.

a) Prove that $(a + b)^2 \equiv (a^2 + b^2) \pmod{2}$

b) Prove that $(a + b)^3 \equiv (a^3 + b^3) \pmod{3}$

c) Prove that it is **not** always true that $(a + b)^4 \equiv (a^4 + b^4) \pmod{4}$